

Markscheme

Mathematics: analysis and approaches
Sample question

1. (a)

Consider the numbers $a = 6.21 \times 10^5$ and $b = 1.48 \times 10^{-6}$. Calculate $C = \sqrt[10]{\left(\frac{a}{b}\right)^3}$, giving your answer correct to 3 significant figures.

attempt to substitute a and b into the expression for C (M1)

$$C = 3068.05 \dots$$

A1

$$C = 3070 \quad (3.07 \times 10^3)$$

A1

Note: Award (M1)A1A0 if the unrounded value 3068 (or similar) is given as the final answer without correct rounding to 3 s.f.

Note: Award (M1)A0A0 if only $\frac{a}{b}$ is calculated correctly with no further correct working towards C .

Note: Do not accept a final answer given in calculator notation, e.g. 3.07E3.

[3 marks]